

RAHUL RAVI VK

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SUMMARY

Instrumentation and controls engineer with 6+ years deploying closed-loop control, PLC/DCS automation, and sensor integration on live industrial plants, now pursuing an M.Eng. in Robotics. Combines hands-on hardware experience (PID tuning, field instrumentation, safety interlocks) with graduate work in ROS 2, perception, and motion planning. Seeking roles that bridge physical automation, instrumentation, and robotic systems.

EDUCATION

M.Eng. in Robotics | University of Maryland, College Park Sep 2025 – May 2027

GPA: 3.9/4.0 | Coursework: Planning and Perception for Autonomous Robots, Control Systems for Robotics, Autonomous Robotics, Robot Modeling, Intro to C++

B.Eng. in Instrumentation & Control | L.D. College of Engineering, Gujarat, India Aug 2014 – May 2018

GPA: 3.3/4.0

WORK EXPERIENCE

Instrumentation & Controls Engineer | KRIBHCO Fertilizers Ltd, Gujarat, India Oct 2018 – Jul 2025

- Designed and tuned closed-loop control on pressure, flow, and level instrumentation across **Honeywell Experion PKS C300 DCS** and **Siemens PLCs**; cut loop overshoot, settling time, and process downtime by **30%** via systematic PID retuning and actuator nonlinearity analysis.
- Built advanced control strategies including pump torque-converter speed control and bistable-relay emulation logic, reducing equipment downtime by up to **50%** and false trips by **90%**.
- Owned full automation lifecycle for safety-critical units: field wiring, sensor calibration, PLC sequencing and interlocks, alarm rationalization, loop checking, FAT/SAT, and operator handover.

PROJECTS

Baron Mobile Robot — Hardware-Up Skid-Steer Platform *Raspberry Pi 4, Arduino, BNO055 IMU, Python*

- Built a skid-steer robot end-to-end from bare hardware: wheel-encoder odometry (10 CPR, 120:1 gearing, 65 mm wheels), **closed-loop PID velocity control**, and BNO055 IMU fusion over serial UART; implemented slip-compensated pivoting, teleop, and CSV data logging for controller identification.

Vision-Based Arrow Detection for Autonomous Navigation *PyTorch, OpenCV, Raspberry Pi*

- Developed arrow-direction classifier for the Baron platform, progressing from classical CV (HSV, PCA) to a **ResNet18** transfer-learning model achieving **100% test accuracy**; deployed inference on Raspberry Pi 4 with live camera feed closing the loop with the motion controller.

Optimal Control for Nonholonomic Mobile Robot (ENPM 667) *MATLAB, Genetic Algorithms, Adaptive PID*

- Reproduced an IEEE Access (2025) paper separating kinematic and dynamic control: GA-optimized kinematic gains (K_ρ , K_α , K_β) over a 6×12 grid of initial postures with closed-loop stability constraints, 2D cubic-spline interpolation, and an **adaptive PID** dynamic controller with gradient-descent gain tuning; verified Lyapunov stability of the adaptive law.

Franka Panda Pick-and-Place *ROS 2 Humble, MoveIt 2, C++*

- Built end-to-end pick-and-place pipeline for a 7-DOF Franka arm using MoveIt 2 motion planning, a custom C++ execution node with named-pose sequencing, collision-aware planning, and gripper control.

UR5 Browser-to-Robot Drawing System *ROS 2, IKPy, WebSockets*

- Designed decoupled architecture bridging a browser drawing interface to a Python IK backend (IKPy) via WebSockets, driving a UR5 in Gazebo with real-time Cartesian-to-joint-space tracking.

Truck Twin-Trailer Simulation & Digital Twin *ROS 2 Control, Gazebo, SolidWorks, Blender*

- Modeled a multi-link articulated truck in SolidWorks, exported to URDF for ROS 2 Control simulation and FBX for Falcon Editor digital-twin deployment; implemented Python proportional controller for point-to-point navigation with LiDAR integration.

TECHNICAL SKILLS

- Industrial Automation & Instrumentation:** Honeywell Experion PKS C300 DCS, Siemens PLC (Ladder Logic), SCADA, safety interlocks, alarm rationalization, loop checking, FAT/SAT, field instrumentation, sensor calibration, 4–20 mA / HART
- Control & Estimation:** PID, Adaptive PID, LQR, LQG, Kalman filtering, state-space modeling, Lyapunov stability, trajectory optimization, closed-loop system identification
- Robotics & Simulation:** ROS 2 (Humble; Actions, Services, Topics, ros2_control), MoveIt 2, Gazebo, URDF/Xacro, RViz2, Falcon Editor, digital twins
- Embedded & Hardware:** Raspberry Pi 4, Arduino Nano, BNO055 IMU, wheel encoders, GPIO, UART/serial, soldering, wiring, benchtop debugging
- Programming & Tools:** C++17, Python (NumPy, SciPy, OpenCV, PyTorch, IKPy, ResNet / transfer learning, MATLAB/Simulink, Bash, WebSockets, SolidWorks, Blender, Linux (Ubuntu), Git, Docker

CERTIFICATIONS

Siemens – Basics of PLC & SCADA | Honeywell – Experion PKS C300 Fundamentals